Explainable AI based Maternal Health Risk Prediction using Machine Learning and Deep Learning

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***Abstract*—The significance of prenatal care is highlighted since effective maternal health care increases the probability of a successful pregnancy and a healthy baby. One of the most crucial aspects of pregnancy is the diagnosis of high risk pregnancy, which can be extremely beneficial to expectant mothers. Early diagnosis can also lower maternal mortality and morbidity. The goal of this study is to apply machine learning and deep learning algorithms to determine the risk level based on pregnancy risk factors. The analysis of risk factors in this study has been conducted using an existing dataset (Maternal Health Risk Data), and a comparison of various machine learning and deep learning algorithms reveals that the Gradient Boosting algorithm provides the highest accuracy in terms of risk level prediction, with an accuracy of 90.640%. Additionally, we have implemented Explainable AI (LIME and SHAP) to discover the exact explanation of the prediction that has been made.**

***Index Terms*—Maternal Health Care, Risk Factors, Machine Learning, Deep Learning, Explainable AI.**

I. Introduction

During pregnancy, in order to ensure the healthy development of the fetus and the safety of delivery, every moment throughout pregnancy should be monitored closely. Because they do not possess sufficient understanding of maternal healthcare throughout and after pregnancy, many pregnant women pass away from pregnancy problems. According to statistics, the maternal mortality rate is higher in rustic and poverty-stricken areas worldwide than in metropolitan and prosperous areas [1]. Finding the fundamental causes, which is crucial for any indication at the beginning, can help to reduce the majority of challenges associated to maternal health. The aim of this research work is to solve the problem of maternal health, particularly in rural areas of a developing nation, by identifying risk factors and letting patients know if they are at high, medium, or low risk. This approach will lessen issues for both expectant mothers and new babies. Therefore, this process increases the safety of the fitness of both the mother and the unborn child.

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Many risk factors need to be measured during pregnancy. Age, Diastolic and Systolic Blood Pressure, Blood Glucose Level, Heart Rate, Body Temperature, Body Mass Index, Blood Oxygen, Physical Activity, First Trimester Nausea, Fetal Heart Rate, Abnormal Fetal Protein, Fetal Motion Activity, etc. are a few of these. The primary priority for maintaining excellent health during pregnancy is timely diagnosis and the appropriate treatment. Early prenatal care and diagnosis are vital in preventing unnecessary maternal fatalities, especially in rural areas.

In low and middle income nations, 99% of maternal deaths worldwide actually occur. On the whole, out of 800 maternal deaths (daily), the majority passed away from critical perinatal hemorrhage, hypertension and risky miscarriage [2]. The ability to anticipate high-risk pregnancies in advance is extremely helpful as it can occasionally assist doctors decide whether to continue or abort the pregnancy, which lowers deaths in expectant mothers [3]. Consequently, the timely identification of maternal high-risk pregnancies can be a successful solution with the aid of a predictive model that uses machine learning and deep learning techniques. And also, we have applied Explainable AI (LIME and SHAP) to find out the actual reason of prediction.

II. Literature Review

Many research have been conducted in the subject of predicting high-risk pregnancies, one of which analyzed various algorithms for maternal high-risk pregnancy detection in 2020 and demonstrated the applicability of two sophisticated algorithms, artificial neural network and convolutional neural network. Data extraction from the input is performed in real time, followed by parsing by a neural network that has been trained and output specification. The final score in this study has a 90% accuracy rate [4].

A study that implements a modified decision tree algorithm to diagnose high-risk pregnancies and is presented in 2020. Data from six hospitals were used in this study, and the data also includes six independent factors and one outcome variable. The study covered the years 2018 to

2020. Additionally, IoT technology was utilized to obtain data for this research. In this investigation, it was found that the modified decision tree algorithm performed 97% more accurately than comparable methods in predicting the model. The implementation also utilized WEKA and Python software [5].

In [6], Poon et al. discuss their work on early detection and prediction of hypertensive problems in the first period of pregnancy in the population of London, United Kingdom. Using logistic regression, a 5% false positive rate and a 90% early detection rate for preeclampsia were established.

Park et al. [7] offer a multiple logistic regression-based algorithm to forecast the risk of preeclampsia in an Australian population. The study correctly recognized preeclampsia in 95% of the women, with a 10% false positive rate.

III. Methodology

After collecting the data, we preprocessed it and applied several machine learning and deep learning algorithms to it to train our model. After that, we evaluated the model and used Explainable AI to explain it. Fig. 1. provides an illustration of the methodology of our system.

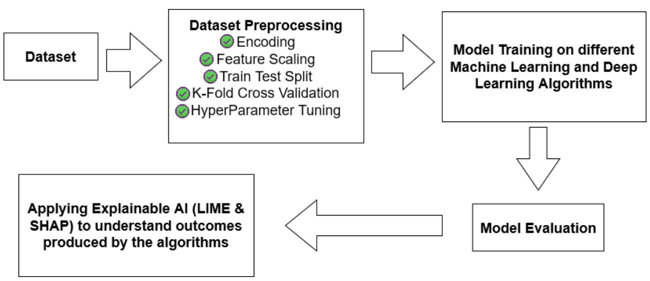

Fig. 1: Overall Working Flow of the Proposed Method

A. Dataset

We have collected this dataset from Kaggle [8]. Data has been collected from various hospitals, community clinics and maternal health centers via the IoT-based risk monitoring system. The dataset includes 1014 records and six features with one level. In Table I, the six independent variables are discussed.

TABLE I: Six Independent Variables Description

Independent Variables	Description	Unit
Age	When a woman is pregnant	years
SystolicBP	Upper value of *BP	mmHg
DiastolicBP	Lower value of *BP	mmHg
BS	Blood glucose levels	mmol/L
BodyTemperature	Human body temperature	fahrenheit
HeartRate	A normal resting heart-rate	beats/minute

*BP indicates Blood Pressure

The one dependent variable is Risk Level. Risk Level indicates risk intensity level that can be predicted during pregnancy considering the six features mentioned above. From the dataset, we obtained three types of risk level: Low Risk, Mid Risk and High Risk. Fig. 2. shows the distribution of the dataset.

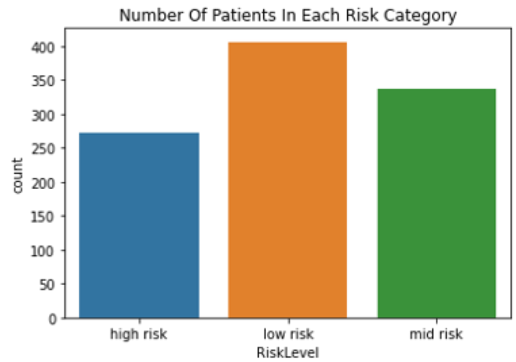

Fig. 2: Risk Level Count

B. Dataset Preprocessing

1) Encoding: At first, we check if our data set contains a null value and we do not find a null value there. This dataset has categorical values only in the Risk Level column. Therefore, we need to pre-process the dataset. For the pre-processing, we have used Ordinal Encoder as the encoding technique. After applying the ordinal encoder, the low, mid, and high risk have become 0.0, 1.0, and 2.0, respectively.

2) Feature Scaling: We have used Standard Scaler as the feature scaling technique. The purpose of using the standard scaler is to standardize the functionality of the input dataset. Standard Scalar normalizes the characteristics of the dataset by scaling to unit variance and removing the mean (optional) using column summary statistics for the training set samples.

3) Train Test Split: The train-test split method has been used to assess the effectiveness of machine learning and deep learning algorithms. At 20% and 80% of the real data, the test set and training set are split respectively. The test set has 203 records, whereas the training set includes 811 records. The test set is solely used to make predictions; the train set is used to fit the model.

4) K-Fold Cross Validation: Using a confined sample of data, machine learning models are often tested using cross validation, a resampling technique. The algorithm only requires one variable, k, which represents the number of groups that will be produced from a given data sample. Because of this, the procedure is also known as k-fold cross-validation. When we chose k = 10, the dataset in our experiment is randomly shuffled and divided into 10 groups. As a result, it is known as the 10-fold cross-validation method. The remaining folds will serve as the test set, while (k-1) folds will serve as the training set. After that, the model is trained on the training set and verified on the test set.

5) HyperParameter Tuning: To get the best model in machine learning, there is something referred to as Hyperparameter Tuning. GridSearchCV is one of the most common approaches. We have used GridSearchCV method as hyperparameter tuning. This is a technique to find the best parameter values from the given set of parameters grid.

C. Model Training

We have applied 14 algorithms to estimate the prediction of our model. There are 14 in total, 13 of which are machine learning algorithms, and one of them is a deep learning algorithm. Table III shows the accuracy of all algorithms used in this system. Some of them are discussed in this section.

1) Random Forest: A supervised machine learning algorithm frequently utilized in classification and regression issues can be Random Forest. It is based on the idea of ensemble learning [9], which joins together several classifiers to address a complicated problem. By creating decision trees for several classifiers and using the majority vote for ranking, it enhances model performance. We get an accuracy of 90.148% from the train-test split method and 84.402% from the k-fold cross validation method.

Each decision tree is constructed using a set of decision rules that recursively segregate the data according to the values of the input features. The impurity of a partition is typically measured using a metric such as Gini impurity or entropy. The formula for Gini impurity is:

$$Gini = 1 - \sum (pi)^2 \tag{1}$$

In equation (1), pi is the proportion of samples in the partition that belong to class i. To build a decision tree, the algorithm selects the feature that best separates the data based on the impurity of the partitions. This is done by calculating the information gain of each feature. The formula for information gain is:

$$IG(S, F) = I(S) - \sum (|Sv|/|S|)I(Sv) \tag{2}$$

In equation (2), S is the current set of samples, F is the feature being considered for splitting, Sv are the subsets of S resulting from the split, —Sv— is the number of samples in Sv, and I(S) is the impurity of set S.

Overall, Random Forest is a powerful and adaptable algorithm that combines the characteristics of several decision trees to increase model correctness and stability.

2) Gradient Boosting: Gradient Boosting is a machine learning algorithm being used regression, classification, and other purposes. It is an ensemble learning method in which numerous weak learners (typically decision trees) are combined to generate a strong learner. Gradient Boosting optimizes the loss function [10] of the model using a gradient descent approach. We get an accuracy of 90.640% from the train-test split method and 84.990% from the k-fold cross validation method. In summary, Gradient Boosting initializes a simple model and iteratively improves upon it by building a weak learner to predict the gradient, updating the model, and repeating the process. This gradual improvement leads to a strong learner with high predictive accuracy.

The loss function quantifies how well the model fits the data. The loss function for regression problems is frequently the mean squared error (MSE), which is defined as:

$$L(y, f(x)) = (y - f(x))^2 \tag{3}$$

In equation (3), y denotes the target variable and f(x) denotes the prediction of the model. The gradient of the loss function in relation to the prediction is calculated as follows:

$$-dL(y, f(x))/df(x) = 2 * (y - f(x)) \tag{4}$$

This measures the direction and magnitude of the error in the current prediction. In equation (4), the negative sign indicates that the gradient points in the opposite direction of the error, so the update to the model should be in the direction of the gradient. A weak learner (often a decision tree) is built to predict the negative gradient (the residual error) of the current model. This is done by minimizing the sum of squared residuals for each leaf node of the tree. The formula for the sum of squared residuals for a leaf node is:

$$\sum (yi - fi)^2 \tag{5}$$

In equation (5), yi is the target variable for sample i, and fi is the prediction of the current model for sample i. The goal is to obtain the values of fi that reduce the sum of squared residuals.

Gradient Boosting is a powerful and versatile technique that leverages the skills of numerous weak learners to build a strong learner capable of fitting complex and non-linear data relationships. To optimize the loss function and steadily enhance the accuracy of the model, the approach employs a gradient descent algorithm.

3) XGBoost: XGBoost stands for Extreme Gradient Boosting which is a distributed and scalable machine learning library with Gradient Boosted Decision Tree (GBDT). XGBoost has been integrated with a variety of other tools and packages such as scikit-learn for Python enthusiasts. From the train-test split method and the k-fold cross validation method, we achieve an accuracy of 89.163% and 81.345% respectively.

Mathematically, XGBoost can be described as follows: Assuming we have a training dataset with n observations and m features, represented as (xi, yi). Here, i=1,...,n, where xi is a vector of feature values and yi is the associated target value. Our goal is to train a f(x) model that can predict the target value y for new instances of x. A model is created by XGBoost as a weighted sum of decision trees. How well the model fits the training data is determined by the loss function, and the negative gradient indicates the direction of steepest descent in the surface of the loss function. By iteratively adding trees to the model, XGBoost minimizes the loss function and improves the accuracy.

Formally, the model is defined as:

$$f(x) = \sum j = 1k.fj(x) \tag{6}$$

In equation (6), fj(x) is the prediction of the jth decision tree on the input vector x, and k is the number of trees in the model. Each tree is determined by the form:

$$fj(x) = wq(x; \theta j) \tag{7}$$

In equation (7), $q(x; \theta j)$ is the decision tree itself, parameterized by θj, and w is the weight of the tree. The weight wj, which is learnt throughout the training process, affects how much each tree contributes to the final prediction. To train the model, XGBoost minimizes the following objective function:

$$obj(\theta) = L(y, f(x)) + \Omega(f) \tag{8}$$

In equation (8), L is the loss function that evaluates the difference between predicted and true values and Ω is a regularization term that penalizes the complexity of the model. The optimization problem is solved using gradient descent, where the gradients are computed using the chain rule of differentiation. The negative gradient is computed as:

$$[\partial L(yi, f(xi))/\partial f(xi)]f'j(xi) \tag{9}$$

In equation (9), $f'j(xi)$ is the gradient of the decision tree with respect to the input xi.

During training, XGBoost uses a process called pruning to avoid overfitting and increase the efficiency of the algorithm. Pruning involves removing branches from the decision tree that do not improve the accuracy of the model on the validation set.

IV. Result Analysis

TABLE II: Accuracy of 14 Algorithms

Serial No	Algorithms	Accuracy	
		Train-Test Split	***K-Fold Cross Validation***
01	KNN	89.163%	85.389%
02	Random Forest	90.148%	84.402%
03	Gradient Boosting	90.640%	84.990%
04	XGBoost	89.655%	81.345%
05	CatBoost	87.192%	81.839%
06	Bagging Classifier	89.163%	82.434%
07	AdaBoost	75.862%	62.713%
08	Decision Tree	88.670%	83.023%
09	Extra Tree Classifier	89.655%	83.032%
10	Logistic Regression	72.906%	64.793%
11	Support Vector Classifier	84.729%	82.338%
12	Gaussian Naive Bayes	71.921%	64.577%
13	Voting Classifier	89.655%	84.011%
14	Multi Layer Perceptron	83.744%	77.708%

As shown in Table II, for all 14 algorithms, the train-test split technique is more accurate than the k-fold cross validation method. As a result, we depict the train-test split accuracy as a bar plot. The accuracy bar plot from the train-test split approach is shown in Fig. 3. Also, we note that the Gradient Boosting Algorithm provides us with the maximum accuracy of 90.640%.

From Table II, we can observe that the accuracy of all algorithms varies significantly. The top-performing algorithms based on the train-test split technique are Random Forest, Gradient Boosting, and XGBOOST, while the least accurate algorithms are Logistic Regression, Gaussian Naive Bayes, and AdaBoost. Additionally, we can see that the k-fold cross-validation technique produces lower accuracy scores for all algorithms compared to the train-test split technique. This could be due to the fact that the k-fold cross-validation technique uses multiple subsets of the data for training and testing, which could result in a more generalized model but with slightly lower accuracy.

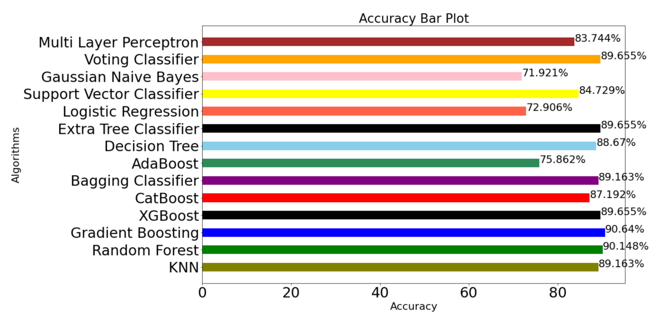

Fig. 3: Train-Test Split Method Accuracy of 14 Algorithms

The bar plot in Fig. 3 provides a visual representation of the accuracy scores for each algorithm using the train-test split technique. It is evident that the top-performing algorithms have accuracy scores above 90%, while the least accurate algorithms have scores below 75%. Finally, we note that the Gradient Boosting algorithm provides the highest accuracy score of 90.64% among all the algorithms tested.

V. Explainable AI

Explainable Artificial Intelligence (XAI) is a procedure that enables us to understand and believe in the outcomes produced by machine learning algorithms. Explainable AI is utilized to express an AI model, its predicted effects, and possible biases. The entire calculation process becomes which is commonly known as a black-box that cannot be interpreted. We have used two famous XAI techniques called LIME (Local Interpretable Model-agnostic Explanations) and SHAP (Shapley Additive Explanations).

A. LIME

Local Interpretable Model-agnostic Explanation (LIME) is a machine learning algorithm that helps to clarify the predictions generated by any black-box model. LIME [11] generates local and interpretable explanations for individual predictions by approximating the black-box model locally using a simpler and more interpretable model. The intuition behind LIME is that it generates a linear model that locally approximates the behavior of the black-box model. LIME achieves [12] this by sampling data points around the point of interest and computing the feature importance for each sample. The prediction of the black-box model is then interpreted using the feature importance weights after fitting a linear model to the data points.

Let x represent the input data point for which we want to provide an explanation for the estimation of the black-box model. This prediction function can be written as f(x). The goal of LIME is to approximate f(x) using a simpler and more interpretable model that can be easily explained. This simpler model is denoted as g(x). To generate the explanation for a specific prediction, LIME selects a set of samples around the data point x. Let D = xi be the set of sampled data points. LIME then weights each sample point by a proximity function that measures how close the sample is to the point x. This proximity function is denoted as d(xi, x), which could be a Gaussian kernel or some other distance metric. LIME then fits a linear model g(x) to the weighted samples D by minimizing the following objective function:

$$L(g) = \sum id(xi, x)[f(xi) - g(xi)]^2 + \Omega(g) \tag{10}$$

In equation (10), the first term estimates the difference between the anticipation of the black-box model and the simpler model, and the second term is a regularization term that prevents the simpler model from overfitting to the data.

$$wj = \partial g(x)/\partial xj \tag{11}$$

These weights show how much every feature influences the simpler prediction in equation (11). These weights are then used by LIME to make the prediction clearer. Generally speaking, when LIME [13] receives a predictive model and a test sample, LIME samples and obtains a surrogate data set and selects features from the surrogate data set. We have implemented three cases here.

Case 01:

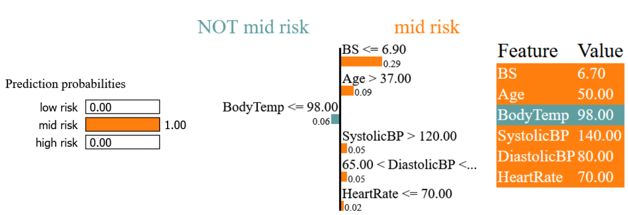

Fig. 4: Prediction of Mid-Risk Level With 100% Confidence

Fig 4. shows that the model predicts this particular patient to have mid risk level with 100% of confidence, and explains the prediction because the blood glucose level (BS) is less than 6.90, age is higher than 37, SystolicBP is more than 120, DiastolicBP is higher than 65, HeartRate is equal to 70. On the right, we can observe the real feature value for the patient.

Case 02:

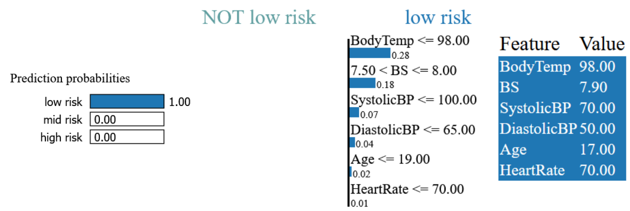

Fig. 5: Prediction of Low-Risk Level With 100% Confidence

Fig 5. shows that the model predicts this specific patient to have low risk level with 100% of confidence, and explains the prediction because the body temperature is equal to 98, the blood glucose level (BS) is less than 8.0, age is lower than 19, SystolicBP is less than 100, DiastolicBP is lower than 65, HeartRate is equal to 70. On the right, we can observe the actual feature value for the patient.

Case 03:

Fig 6. illustrates that the model predicts this particular patient to have high risk level with 100% of confidence, and explains the prediction because the blood glucose level (BS) is greater than 8.0, age is higher than 25, DiastolicBP is greater than 65, HeartRate is higher than 76. On the right, we can observe the real feature value for the patient.

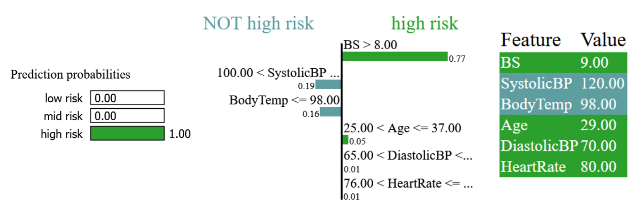

Fig. 6: Prediction of High-Risk Level With 100% Confidence

B. SHAP

SHAP is the abbreviation of SHapley Additive exPlanations. By determining the impact of each feature to the prediction, this approach [14] seeks to explain how an instance has predicted. Summary plot is one of the visualization tools available in SHAP for making models understandable.

Mathematically, the SHAP value for a feature xi in a prediction y can be defined as follows:

$$SHAP(xi) = \sum (f(z \cup \{xi\}) - f(z))/S \tag{12}$$

In equation (12), f is the machine learning model, z is a subset of the features that have already been included in the explanation, $z \cup \{xi\}$ is the set of features that includes xi, and S is the number of subsets that include xi.

Intuitively, the SHAP value for a feature xi measures the difference between the prediction for the current set of features z and the prediction when xi is added to z. SHAP employs a weighted linear regression model that approximates the black-box model regionally to compute the Shapley values of all features. The Shapley values determine the weights of the linear model, while the inputs to the model are the attributes of the data point x. The results provided by the black-box model is the output.

A summary plot or a dependence plot can be used to illustrate the SHAP values, which demonstrate the contribution of each feature to the prediction for individual data points. Overall, SHAP is a powerful and flexible algorithm that provides interpretable explanations for complex machine learning models.

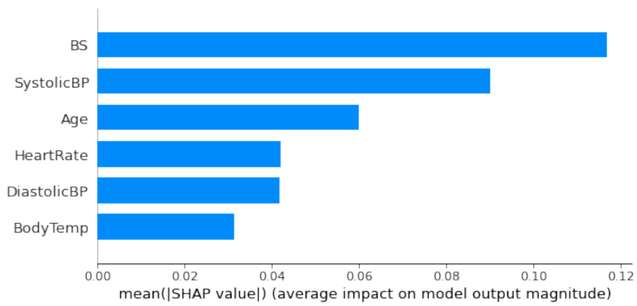

Fig. 7: Summary Plot For Random Forest

From most significant to least important, Fig. 7 lists the key characteristics for the Random Forest algorithm. According to the model, Blood Glucose Level (BS) has the greatest predictive value. The second feature with the best predictive ability is Systolic Blood Pressure. The third factor with the highest predictive value is age.

Fig. 8 lists the key features for the XGBoost algorithm from most significant to least significant,. According to the model, Systolic Blood Pressure has the highest strength for prediction. The second feature with the best predictive potential is Blood Glucose Level (BS). The third factor with the highest predictive value is age.

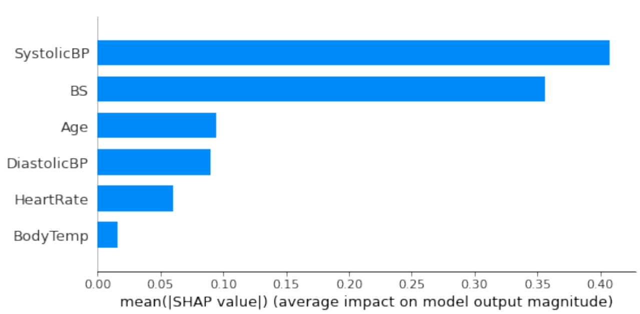

Fig. 8: Summary Plot For XGBoost

Finally, we can say that Blood Glucose Level (BS) and SystolicBP both are two discriminative features.

VI. Conclusion

In conclusion, it should be noted that Artificial Intelligence is permeating virtually every industry, and maternal health is no different. The risk level of an existing dataset has classified and predicted using machine learning and deep learning methods. This artificial intelligence immediately produces results, but its outcome cannot be fully relied upon; it is necessary to understand the real cause. For this reason, we have used Explainable AI. The key benefit of interpretable machine learning and deep learning applications is that the output is objective because it is based on actual data and outcomes and identifies the most important characteristics for physicians. Future study on maternal and fetal health, immunology, and other aspects will be included to the extensive analysis. In order to develop a hybrid algorithm or a new algorithm that can deliver the best results, we will attempt to evaluate large amounts of data.

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